[INCH-POUND]
OO-R-2898
March 14, 1997
SUPERSEDING
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22 October 1987

FEDERAL SPECIFICATION

REFRIGERATORS, MECHANICAL, STORAGE STAND (FOR BULK MILK DISPENSERS)

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers refrigerated storage stands designed to store filled bulk milk containers and to serve as a counter-height stand for supporting a bulk milk dispenser.

1.2 Classification. Refrigerated storage stands shall be of the following sizes and styles, as specified (see 6.2):

Sizes

Size 2-5 - Storage capacity for two 5-gallon (18.9 liters (L)) milk cans or containers.
Size 3-5 - Storage capacity for three 5-gallon (18.9 L) milk cans or containers.

Styles

Style A - Leg mounted.
Style B - Caster-mounted.
Style C - Mounted for shipboard use.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 1581), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4110

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification

OO-D-450 - Dispensers, Bulk Milk, Mechanically Cooled (and Containers, Milk).

Military Standards

MIL-STD-167/1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility.

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MIL-STD-462 - Department of Defense Test Method Standard for Measurement of Electromagnetic Interference Characteristics.

(Copies of federal and military specifications, standards, and handbooks required by contractors in connection with specific procurement functions are obtained from Defense Automated Printing Services, Attn: DoDSSP, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE-STD-23-78.
ASHRAE-STD-34-1992.

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society of Heating, Refrigerating, and Air-Conditioning, 1791 Tullie Circle, NE, Atlanta, GA 30329.)

ASTM

ASTM A 167 - Stainless and Heat-resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
ASTM A 176 - Stainless and Heat-resisting Chromium Steel Plate, Sheet, and Strip.

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(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society for Testing Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

NATIONAL SANITATION FOUNDATION (NSF)

NSF No. 7 - Food Service Refrigerators and Food Service Storage Freezers.
NSF Listing of Food Service Equipment.

(Private sector and civil agencies may purchase copies of these voluntary standards from the National Sanitation Foundation, 3475 Plymouth Road, P.O. Box 1468, Ann Arbor, MI 48106.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 471 - Commercial Refrigerators and Freezers.
UL Electrical Appliance and Utilization Equipment Directory.

(Private sector and civil agencies may purchase copies of these voluntary standards from the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Defense Automated Printing Services, Attn: DoDSSP, Building 4D, 700 Robbins Avenue, Philadelphia, PA, 19111-5094).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The refrigerated storage stands shall consist essentially of an insulated storage compartment having a front-mounted, insulated door and top designed specifically to support a fully-loaded bulk milk dispenser conforming to either type I or type II of OO-D-450. The size 2-5 stand shall support a size 2-5 dispenser; the size 3-5 stand shall support a size 3-5 dispenser. The stand shall include a complete refrigeration system with automatic controls necessary to maintain the operating temperatures specified herein.

3.2 Standard commercial product. The refrigerated storage stands shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the refrigerated storage stands being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market

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through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.4 Codes and standards. The refrigerated storage stand shall conform to the requirements of UL 471 and NSF No. 7, as applicable.

3.4.1 Applicability. Details for equipment construction, design, and performance covered by this specification shall be supplementary to the requirements of the standards cited in 3.4 and compliance with and approval under the UL or NSF standard shall not relieve the supplier of the responsibility for meeting the requirements of this specification.

3.4.2 UL certification. Acceptable evidence of meeting the requirements of UL 471 shall be the UL certification symbol or label, listing of the model number in the UL Electrical Appliance and Utilization Equipment Directory, or a certified test report (see 6.2), from a recognized independent testing laboratory indicating the storage stand has been tested and conforms to UL 471. Such evidence must be acceptable to the contracting officer.

3.4.3 NSF certification. Acceptable evidence of meeting the requirements of NSF No. 7 shall be the NSF certification symbol or label, listing in the current edition of the NSF Listing of Food Service Equipment, or a certified test report (see 6.2), from a recognized independent testing laboratory indicating the refrigerated storage stand has been tested and conforms to NSF No. 7. Such evidence must be acceptable to the contracting officer.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.5.1 Corrosion-resistant steel. Corrosion-resistant (stainless) steel specified herein for construction of the exterior shell and inner liner shall be any class 302 or 304 of ASTM A 167 or class 430 of ASTM A 176.

3.5.2 Insulation. Insulation shall be in accordance with NSF No. 7. Insulation shall be of a material and so installed that compaction, settlement, or separation of the insulation will not occur under normal conditions of shipment and operation. Insulation shall be closed and sealed to

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afford protection against condensation, spillage, and seepage. Insulation shall be vapor-proofed to prevent undue migration of moisture vapor. The thickness of insulation shall be sufficient to insure compliance with all performance requirements of 3.7.

3.5.3 Refrigerant. The refrigerants shall be a non ozone depleting type, and shall fall in ASHRAE safety group "A1" per ASHRAE-STD-34-1992.

3.6 Design. The refrigerated storage stands shall be designed and constructed to rigidly support, on the top, a fully-loaded bulk milk dispenser of the applicable type and size specified in 3.1 and simultaneously support, in the storage compartment, a full complement of bulk milk containers. The dimensions of the units shall be as specified in table I. The design of the refrigerated storage stands shall be predicated on the size, weight, and loading impact of filled, 5-gallon (18.9 L), dispenser-type milk cans conforming to MIL-C-22785. The refrigerated storage stands shall also be suitable for use with the plastic, single-service and multiple-service bulk milk containers specified in OO-D-450.

TABLE I. Dimensions (inches).

Refrigerator Size	Acceptable range of exterior dimensions			Minimum interior dimensions		
	Height	Width	Depth	Height	Width	Depth
2-5	34-36 inches (86.36 - 914.4 mm)	27-30 inches (685.8 - 762 mm)	24-25 inches (609.6 - 635 mm)	22 inches (558.8 mm)	22 inches (558.8 mm)	11 inches (279.4 mm)
3-5	34-36 inches (863.6 - 914.4 mm)	38-40 inches (965.2 - 1 016 mm)	24-25 inches (609.6 - 635 mm)	22 inches (558.8 mm)	32 inches (812.8 mm)	11 inches (279.4 mm)

3.7 Performance. The refrigerated storage stands shall meet the performance requirements of the National Sanitation Foundation for the size and style specified (see 6.2).

3.8 Construction. The storage compartment shall consist of an outer shell, inner liner, framing, door, hardware, and a leg- or caster-mounting. The outer shell and inner liner shall enclose insulation meeting the requirements of 3.5.2.

3.8.1 Outer shell. The outer shell for the storage compartment shall be fabricated or formed of corrosion-resistant steel of the same thickness as used on models approved under the standards cited in 3.4. Removable panels or covers shall be provided as required to provide ready access for servicing of the refrigeration system.

3.8.2 Inner liner. The walls and top of the inner liner shall be fabricated or formed of corrosion-resistant steel. The liner bottom shall also be corrosion-resistant steel. The liner bottom shall be formed and braced to support a full complement of fully-loaded bulk milk cans, or containers, and to withstand without permanent deflection or distortion the impact normally associated with the

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loading operation. The liner bottom shall be connected to the liner walls so that no liquid can leak into the insulation. This connection shall be able to withstand the impact of the loading operation.

3.8.3 Framing. Framing shall be required to support the dispenser and the fully-loaded storage compartment, and to house the components of the refrigeration system. The framing shall consist of structural steel shapes or plates. The steel framing shall be plated or treated and painted to resist corrosion in accordance with the manufacturer's standard practice. When specified (see 6.2) a foam insulating cabinet structure that meets strength requirements of 3.1 shall be used.

3.8.4 Door. The door shall consist of an inner and outer liner of the material and thickness specified for the storage compartment's liner and shell. Insulation shall be as specified in 3.5.2. The door shall be fitted with an NSF-approved gasket. Door framing shall include suitable plastic or synthetic rubber breaker strips to minimize conduction between the compartment liner and shell. The door shall be hinged on the right unless left side hinges are specified (see 6.2). A thermometer conforming to NSF No. 7 shall be mounted on the door.

3.8.5 Hardware. The door latch, striker plate, hinges, and other miscellaneous hardware shall be fabricated of corrosion-resistant steel or of chromium-plated metal. Except for copper alloys, chromium-plating shall be applied over a base coating of copper followed by a nickel coating.

3.8.6 Mounting. The refrigerated storage stands shall be mounted on legs or on casters in accordance with the style specified. Two of the four casters on the style B units shall be equipped with foot-operated locking brakes. Legs, feet, and casters shall conform to the applicable requirements of NSF No. 7. For shipboard use, legs shall be constructed of corrosion-resistant steel and adequately constructed to permit secure attachment to mounting pad or steel deck.

3.9 Refrigeration system. The refrigeration system shall consist essentially of a hermetic motor-compressor unit, an air-cooled condenser, a drier, evaporator sections with capillary tube or thermal expansion valve, and an automatic thermostat control. Design, construction, and rating of the system shall be in accordance with ASHRAE-STD-23-78. The condensing unit shall have an application rating (not a standard rating) to meet the performance requirements of 3.7 at the specified ambient temperatures. Refrigeration systems shall be completely dehydrated, fully charged, and tested for leaks prior to shipment. Unless otherwise specified (see 6.2), the refrigeration system shall be designed for operation on normal 120-volt, 60 hertz, alternating current. When a low pressure sensing switches required, (see 6.2) it shall be of the manual reset type.

3.10 Identification marking. Identification shall be permanently and legibly marked directly on the refrigerated storage stand or on a corrosion-resisting metal plate securely attached to the storage stand at the source of manufacture. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.11 Finish. The exterior cabinet surfaces and the interior compartment surfaces shall be finished in accordance with NSF No. 7.

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3.12 Spare parts. When specified (see 6.2), spare parts shall be furnished and shipped with each unit. The spare parts required and the quantity thereof shall be as specified in the contract, or, when applicable, shall be determined in accordance with the provisioning procedures of the contract.

3.13 Electromagnetic compatibility. When specified for shipboard use (see 6.2), the refrigerated storage stands shall be designed and equipped for electromagnetic compatibility in accordance with the requirements of MIL-STD-461. The equipment shall meet the emission and susceptibility requirements for CE101, CE102, and RE102 (see 4.3.1).

3.14 Inclined operation. When specified for shipboard use (see 6.2), the refrigerated storage stands shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other, with no spillage of products when tested as specified in (see 4.3.2).

3.15 Environmental suitability. When specified for shipboard use (see 6.2), the refrigerated storage stands shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness or damage when tested as specified in (see 4.3.3).

3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.16.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.4 Solder. All surfaces to be soldered (when applicable) shall be thoroughly cleaned prior to soldering. Solder joints shall be smooth with no pin holes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as

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specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.4.2 and 3.4.3. The Government reserves the right to examine and test all storage stands to determine the validity of the certification.

4.3 Quality conformance tests.

4.3.1 Electromagnetic interference test. When required for shipboard use (see 6.2), each refrigerated storage stand shall be tested by the contractor in accordance with test methods CE101, CE102, and RE02 of MIL-STD-462. The contractor shall furnish written certification that the equipment meets the requirements of MIL-STD-461.

4.3.2 Inclined operational test. When required for shipboard use (see 6.2), the refrigerated storage stand shall be positioned with the base set at an angle of 15 degrees, then operated for 30 seconds at each side of the vertical planes at right angles to each other. At each of these positions, observe for the conformance with the requirements of 3.14. For the test, the unit shall be loaded with bulk milk containers.

4.3.3 Environmental test. When required for shipboard use (see 6.2), the refrigerated storage stand, under normal conditions, shall be tested in accordance with MIL-STD-167/1, type I equipment. The refrigerated storage stand shall be secured to the test machine in the same manner that it will be secured on shipboard. Failure to meet the requirements of 3.15 shall be cause for rejection.

4.3.4 Operational test. Each end item shall be operated to determine compliance with the requirements specified herein. The test shall be conducted at ambient room temperatures

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provided the ambient temperature is 65 °F (18.3 °C) or more. Any of the following defects shall be cause for rejection:

a. Failure of refrigerator to start (see 3.9).
b. Current draw above maximum nameplate rating at rated voltage (see 3.7 and 3.9).
c. Failure of door thermometer to operate (see 3.8.4).
d. Leaks in the refrigeration system (see 3.9).

5. PACKAGING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The refrigerated storage stands are intended for use with bulk milk dispensers in dining facilities, cafeterias, hospitals, soda-fountains, and any other installations where milk is to be refrigerated and dispensed in sanitary manner.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size and style required (see 1.2).
c. When certified test report is required (see 3.4.2 and 3.4.3).
d. When door shall be hinged on the left instead of the right (see 3.8.4).
e. When refrigeration system shall be designed for operation on a power supply other than as specified (see 3.9).
f. When a foam insulation cabinet is acceptable in lieu of framing (see 3.8.3).
g. When a low pressure sensing switch is required (see 3.9).
h. When spare parts are required and the parts and quantities to be furnished (see 3.12).
i. When units require electromagnetic testing for shipboard use (see 3.13 and 4.3.1).
j. When units require inclined testing for shipboard use (see 3.14 and 4.3.2).
k. When units require environmental testing for shipboard use (see 3.15 and 4.3.3).

6.3 Supersession data. This specification replaces Military Specification MIL-R-82047B, dated 22 October 1987.

6.4 Classification cross reference. Classifications used in this specification (see 1.2) are identical to those found in the superseded Military Specification MIL-R-82047B.

6.5 Part or Identifying Numbers (PINs). The specification number, size, and style are combined to form PINs for refrigerated storage stands covered by this document (see 1.2). PINs for the bulk milk dispensers are established as follows:

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PIN: OOR2898 2-5 A

Federal Specification Number

Size - 2-5 two 5-gallon (18.9 L) cans or containers or
Size - 3-5 three 5-gallon (18.9 L) cans or containers

Style A - Leg mounted or
Style B - Caster-mounted or
Style C - Mounted for shipboard use

6.6 Subject term (key word) listing.

Bulk milk dispenser
Food service equipment
Refrigerated storage stand

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to previous issue.

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - YD1
Air Force - 99

Review Activities:
Air Force - 82
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Navy-YD1

(Project 4110-0513)

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